Adaptive Focus Studio - Project One Report

Team Boomers | CSC 6360-002 Mobile App Development | Graduate Pathway
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1. Project Overview

Adaptive Focus Studio is an offline-first Flutter study-session application. It helps students create reusable focus presets, run focus sessions, review completed-session history and local insights, save settings locally, export a session summary, receive a local completion notification, and optionally use Session Scout for an explainable duration suggestion.

2. Problem and Goal

Starting a timer is easy, but planning study sessions consistently and learning from recent study behavior is harder. The project goal was to provide a lightweight focus workflow that does not require an account, cloud backend, or internet connection for its core functions.

3. Application Flow

The application has five main screen roles: Dashboard / Preset Catalog, Create / Edit Preset, Focus Session, History & Insights, and Settings. The Dashboard loads saved presets and Session Scout. The editor validates and saves preset data. The Focus Session screen runs the timer. History displays completed SessionLog records and insights. Settings stores lightweight preferences.

4. Architecture

The project separates the UI, Riverpod state and dependency providers, repositories/services, and local persistence. Widgets do not directly own SQL queries. Repository boundaries keep database behavior separate from presentation code. Riverpod manages the focus timer and repository dependencies across screens.

5. Local Data

SQLite stores the two structured entities. FocusPreset contains id, label, moodTag, and plannedSeconds. SessionLog contains id, presetId, startedAt, completedSeconds, moodTagSnapshot, status, source, and note. SharedPreferences stores default duration, Session Reminders, and Sound. SQLite was chosen for structured records and queries, while SharedPreferences fits small key/value settings.

6. Session Scout

Session Scout is a local rule-based/statistical helper, not a trained machine-learning model. For a selected mood, it uses the three most recent finished matching sessions, converts completed seconds to minutes, takes the median, and clamps the result to 10 through 60 minutes. Fewer than three valid matching sessions produces insufficient-history feedback. Applying a suggestion only prefills the preset editor; it does not auto-save or auto-start a session.

7. Session Scout Held-Out Evaluation

The Graduate evaluation used five fixed synthetic held-out cases. These are evaluation fixtures, not real-user results. The fixed comparison baseline is 25 minutes.

Case	Prior sessions	Actual	Scout	Baseline	Scout error	Baseline error
1	18, 22, 20	21	20	25	1	4
2	32, 38, 35	36	35	25	1	11
3	42, 50, 46	44	46	25	2	19
4	12, 16, 14	15	14	25	1	10
5	25, 30, 28	27	28	25	1	2

Session Scout total absolute error: 6 minutes. Session Scout MAE: 1.20 minutes. Fixed 25-minute baseline total absolute error: 46 minutes. Baseline MAE: 9.20 minutes. Session Scout had lower absolute error in all five synthetic evaluation cases. This small synthetic evaluation does not establish real-world predictive accuracy.

8. Graduate Features

Graduate-pathway features include Riverpod state management and dependency injection, repository architecture, background History insight calculation using Flutter compute(), local notifications, local file export, implicit and explicit animations, unit/widget/integration tests, and the held-out Session Scout comparison.

9. Testing and Verification

Final verification includes flutter analyze with no issues, 50 automated tests passing, measured code coverage of 30.07%, an integration test, release APK testing on Android, and separate second-device Android evidence. The project does not claim the optional greater-than-80-percent coverage bonus.

10. Device Testing

First-device physical Android testing verified the release APK, core five-screen workflow, preset operations, timer flow, History & Insights, Session Scout, settings persistence, local notification, local export, and offline operation. Separate second-device evidence from Parsh supports installation/launch, validation, preset creation, timer/history behavior, settings configuration, Session Scout, local export, and access to stored local data while airplane mode is visible.

11. Contributions

Harshit Jain: 58 commits. Main areas include UI, Riverpod and timer integration, Session Scout, advanced features, integration, documentation, and presentation evidence. Representative Session Scout trace: 4c3264d, f610b77, d1cc89e.

Parsh Jadon: 28 commits. Main areas include SQLite, repositories, CRUD, queries, constraints, and data-layer testing. Representative SessionLog/data trace: d731e3d, 5ad8e97, a98d619, a769cf2.

12. Limitations and Next Iteration

The application is intentionally local-only and therefore does not provide account-based cross-device synchronization. Session Scout uses only three recent finished matching sessions, which may sometimes be unrepresentative. The held-out comparison is small and synthetic. A future iteration should compare longer history windows using a larger set of real usage data while preserving local computation, explanation, and manual user override.

13. Project Links

GitHub Repository: https://github.com/harshitjain25/adaptive-focus-studio
ClickUp Board: https://app.clickup.com/90141778423/v/l/li/901422381505
Final YouTube Presentation: https://youtu.be/7WOXboAY480